

UPPER-BODY CONTEXT NODE (CHEST DEVICE, RING, OR WRISTBAND) FOR CARDIORESPIRATORY, AUTONOMIC, THERMAL, AND MOTION CONTEXT TO CONDITION PHYSIOLOGY INTERPRETATION (NON-LIMITING)

A. Technical Field

This section relates to an upper-body context device configured to capture physiologic context signals (cardiac, autonomic, thermal, respiratory, and motion) used to condition interpretation of other node signals, improve validity gating, and support privacy-preserving cross-device correlation under consent.

B. Summary (Non-Limiting)

In various embodiments, Node 5 comprises an upper-body wearable implemented as a chest-mounted device, ring, wristband, or equivalent form factor. Node 5 acquires one or more modalities including cardiac electrical sensing (ECG electrodes), heart rate variability (HRV) features, optical pulse sensing (PPG), inertial sensing (6-axis IMU), respiratory dynamics (derived from motion, impedance, or other sensing), temperature/thermal context, electrodermal activity (EDA) / skin conductance, and contact integrity/wear detection. A computing system derives context tags and confidence indicators (e.g., motion state, rest state, stress/activation state, thermal coupling state, and wear integrity) and uses these to gate windows, improve artifact rejection, and condition correlation outputs across nodes.

C. Exemplary Form Factors (Non-Limiting)

In various embodiments, Node 5 is implemented as: 1. a chest-mounted device configured for stable ECG and respiration-adjacent motion capture; 2. a ring configured for PPG and motion capture with wear detection; 3. a wristband configured for PPG, EDA, temperature, and motion capture; and/or 4. other upper-body wearables providing substantially similar context signals. Any form factor may be used provided the functional intent of capturing upper-body context for conditioning other node interpretation is maintained.

D. Sensor Suite (Non-Limiting)

In various embodiments, Node 5 includes one or more sensors including, without limitation: 1. ECG electrodes configured to acquire cardiac electrical signals and derive HR/HRV features; 2. PPG sensor(s) configured to acquire optical pulse signals and derive pulse features; 3. 6-axis IMU configured for motion/orientation and activity classification; 4. respiratory dynamics sensing, including respiration derived from chest motion, inertial patterns, impedance, or equivalent methods; 5. temperature sensor(s) configured for thermal context and coupling plausibility; 6. EDA/skin conductance sensing configured to derive autonomic activation features; and 7. contact integrity and wear detection sensing, including electrode contact checks, optical coupling checks, and/or capacitive proximity.

E. Derived Context Features and Tags (Non-Limiting)

In various embodiments, the computing system derives context features including: 1. motion state classifications (e.g., still, light movement, active); 2. autonomic activation indicators derived from HRV and/or EDA features; 3. respiration rate and respiration variability features; 4. thermal context tags and coupling stability indicators; and 5. wear integrity/confidence indicators. In certain embodiments, these context features are computed continuously and summarized into privacy-preserving indices and tags

rather than raw waveforms.

F. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 5 provides user-facing context summaries (rest/activity, stress/activation, recovery, thermal stability, and wear integrity) without requiring other nodes. In optional multi-node embodiments, Node 5 context tags are used to: 1. gate valid windows for other nodes (e.g., exclude high-motion periods); 2. time-align windows across devices under consent; and 3. condition cross-device correlation scores with confidence weighting.